

First name:

Last name:

Student ID:

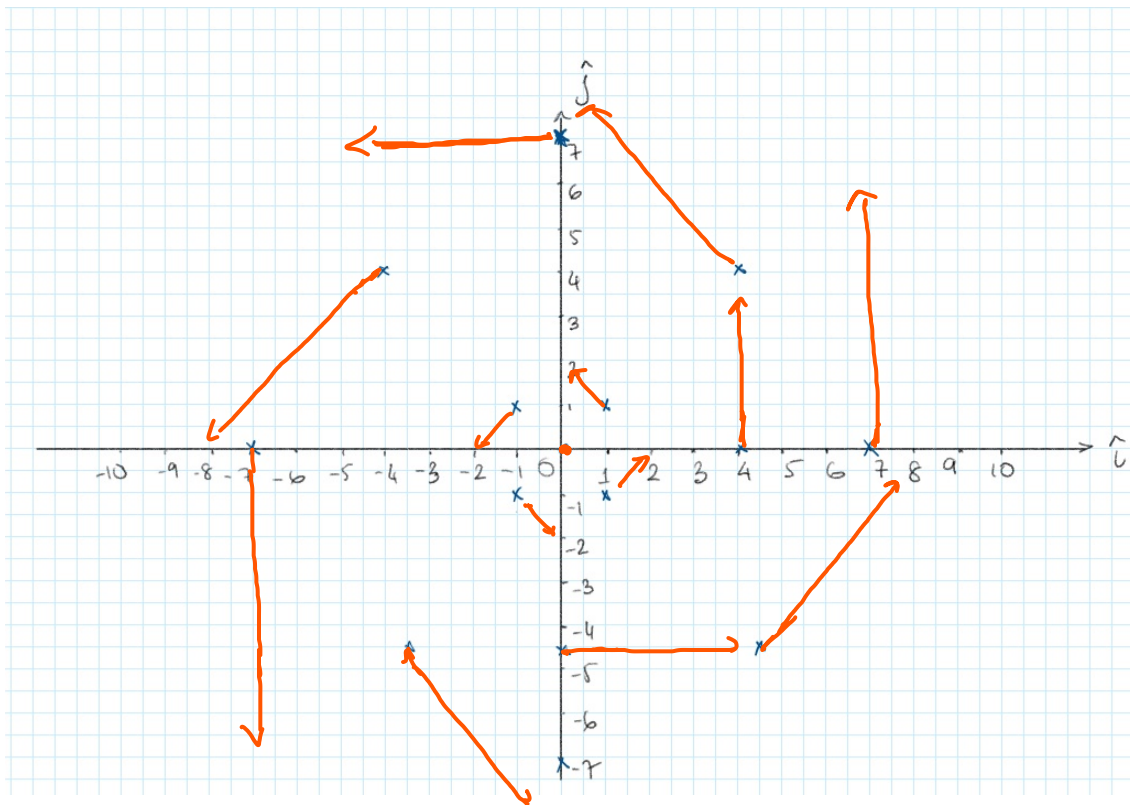
Solutions

Problem 1. Sketch vector arrows at the points marked with 'x' to depict the vector field

$$\vec{v} = -y\hat{i} + x\hat{j}.$$

[5 marks]

Solution.



What I am looking for:

- (1) Arrows all point in the correct-ish direction
- (2) Length of arrows increase as the points get farther away from (0,0)

□

MATH73235
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Spring 2025

Quiz 1

Problem 2. Find the first and second derivatives (with respect to t) of

$$\vec{r}(t) = \begin{bmatrix} 3 \cos(2t) \\ 3 \sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

What I'm look for:

- correct derivatives of $\sin(\cdot)$ and $\cos(\cdot)$
- using Chain Rule on $2t$
- correct derivative of $4t$

□

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Problem 3. Find a parametric representation of a circle in the yz -plane, with center $(0, 4, 0)$ and passing through $(0, 0, 3)$. Sketch the curve on the given axes.
[5 marks]

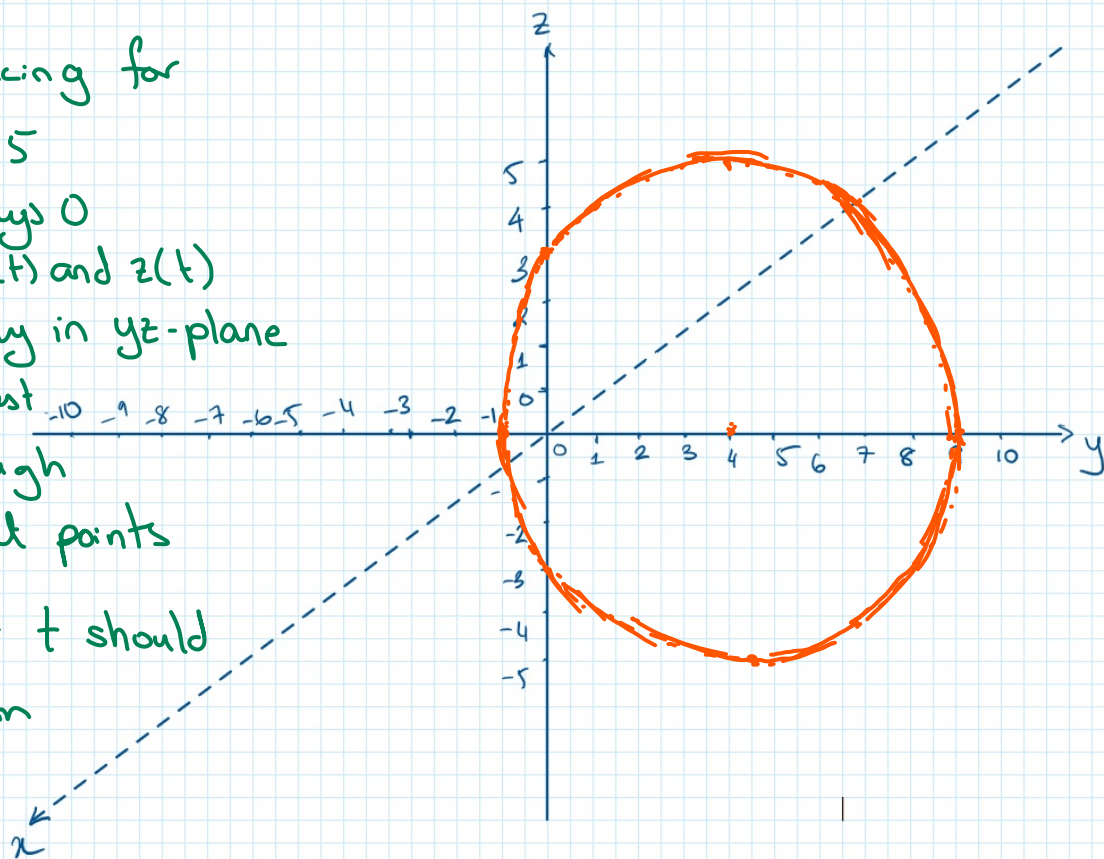
Solution. Parametric representation:

$$\begin{aligned} \text{Circle center: } (0, 4, 0) \text{ , radius} &= \text{distance}((0, 4, 0), (0, 0, 3)) \\ &= \sqrt{0^2 + 4^2 + (-3)^2} = 5 \end{aligned}$$

$$\begin{aligned} \text{Parametric repn: } x(t) &= 0 \\ y(t) &= 5 \cos t + 4 \\ z(t) &= 5 \sin t \end{aligned} \quad 0 \leq t < 2\pi$$

what I'm looking for

- radius of 5
- x is always 0
- correct $y(t)$ and $z(t)$
- circle only in yz -plane
- circle must go through the right points
- parameter t should range from 0 to 2π



Sketch:

